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Dear Editor,

I am submitting our manuscript, “Revisiting the Student Sectioning Problem through Graph Representation Learning,” for consideration for publication in the *Journal of Computer Science and Technology*.

Student sectioning is a classical problem in university course timetabling. We investigate whether the relational structure implicit in student–course enrollments can provide a more informative representation for student grouping. We construct a weighted student co-enrollment graph and apply DeepWalk to learn low-dimensional student representations, referred to as *DeepWalk-SSP*. The learned representations are then supplied to standard clustering algorithms without modifying the clustering stage.

A central aspect of the study is a controlled comparison with both conventional representation and alternative graph/dimensionality-reduction approaches. On six real-world university courses involving 341 students, DeepWalk-SSP achieves an average Silhouette Score of 0.579, compared with 0.460 for PCA+KMeans, 0.153 for the conventional enrollment representation, and 0.151 for spectral clustering. These results indicate that graph representation learning provides an improvement beyond the benefit of dimensionality reduction alone.

The manuscript further evaluates multiple clustering algorithms, embedding and DeepWalk parameters, clustering stability, statistical significance, additional clustering metrics, visualization, and computational cost. The proposed pipeline remains computationally lightweight for the datasets considered.

The work is closely related to recent graph-learning research published in the journal, including network embedding [1], representation learning for structured data [2], and graph convolutional networks for graph-based anomaly detection [3]. Our study complements these directions by investigating graph representation learning in an educational data-mining application and by examining whether learned graph representations can improve a classical clustering-based problem.

We believe the manuscript fits the journal’s scope at the intersection of graph learning, representation learning, clustering, and applied machine learning.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. The source code and datasets used in this paper are available as supplementary material for reviewers.

Thank you for considering our manuscript.

Sincerely,

Mahmood Amintoosi

References

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 - [2] Songjie Niu and Dongyan Zhou. Soop: Efficient distributed graph computation supporting second-order random walks. *Journal of Computer Science and Technology*, 36(5):985–1001, 2021.
 - [3] Jun Yang, Yi-Qiang Sheng, Jin-Lin Wang, and Hong Ni. Cagcn: Centrality-aware graph convolution network for anomaly detection in industrial control systems. *Journal of Computer Science and Technology*, 39(4):967–983, 2024.
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